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BATTERY MANAGEMENT SYSTEM AND BATTERY PACK

Abstract

Provided are a battery management system and a battery pack. The battery pack includes a battery management system. The battery management system includes a first circuit board, a second circuit board, a copper bar and a fastening assembly. The first circuit board and the second circuit board are arranged at intervals and are fixed within a box body. The copper bar is located between the first circuit board and the second circuit board. The fastening assembly includes an insulating member and a first fastener. The copper bar is disposed to be attached to the first circuit board. A first end of the insulating member is connected to the second circuit board, and a second end of the insulating member supports the copper bar. The first fastener is fastenably connected to the insulating member and abuts against a side of the first circuit board away from the second circuit board

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority to Chinese Patent Application No. 202420446586.3, filed with the China National Intellectual Property Administration (CNIPA) on Mar. 7, 2024, the disclosure of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present application relates to the technical field of batteries, for example, a battery management system and a battery pack.

BACKGROUND

[0003] A battery management system (BMS) is an important component of a battery pack. The battery management system is disposed within a box body of the battery pack, and generally includes two circuit boards arranged at intervals in a vertical direction and a copper bar disposed between the two circuit boards. One end of the copper bar is electrically connected to the battery module within the battery pack, and the other end of the copper bar is generally located between the two circuit boards.

[0004] In the related art, in order to fix the copper bar closely to a circuit board on an upper layer to achieve the electrical connection, a common practice is as follows: a screw post is integrally formed on the box body, the screw post passes through a circuit board on a lower side and supports the copper bar, then a threaded hole is extruded on the screw post by using the tapping process, and the screw is fastened in the threaded hole, to fasten the copper bar on the circuit board on the upper layer.

[0005] In the above-described related art, the box body and the screw post are made of the metal, and the copper bar is in contact with the screw post; therefore the current on the copper bar flows to the box body through the screw post, so that the box body has an electric potential. However, the risk of short circuits may occur for the box body with the electric potential, and the risk of electric shocks may occur for an operator. Therefore, the insulation between the box body and the copper bar will be achieved subsequently by using an electrophoresis process, however, the electrophoresis process has the high cost and the complicated process, thereby resulting in the higher manufacturing cost of the product.

SUMMARY

[0006] An embodiment of the present application provides a battery management system disposed within the box body. The battery management system includes a first circuit board and a second circuit board, a copper bar and a fastening assembly. The first circuit board and the second circuit board are arranged at intervals and fixed within the box body. The copper bar is located between the first circuit board and the second circuit board. The fastening assembly includes an insulating member and a first fastener. The copper bar is disposed to be attached to the first circuit board. A first end of the insulating member is connected to the second circuit board, and a second end of the insulating member supports the copper bar. The first fastener is fastenably connected to the insulating member and abuts against a side of the first circuit board away from the second circuit board, to fix the copper bar on the first circuit board.

[0007] An embodiment of the present application further provides a battery pack. The battery pack includes a box body, a battery module and the battery management system described in any of the embodiments. The battery module and the battery management system are disposed within the box body, and the battery module is electrically connected to the battery management system.

[0008] Some embodiments of the present application provide a battery management system. In this battery management system, the first circuit board and the second circuit board are arranged at intervals and are fixed within the box body, the copper bar is supported by the insulating member connected to the second circuit board, and the first fastener is fastenably connected to the insulating member and abuts against the side of the first circuit board away from the second circuit board, so that the copper bar is attached and fixed to the first circuit board, and thus the copper bar is fastenably connected to the first circuit board through the fastening assembly. The fastening assembly of the above-described form is provided so that the copper bar can be insulated from the box body while the copper bar is fastened, and the first fastener is not in contact with the box body; therefore, the copper bar may not form an electrical connection with the box body through the first fastener, thereby insulating and protecting the box body without using the electrophoresis process, and reducing the manufacturing cost.

[0009] According to the battery pack provided in some embodiments of the present application, the manufacturing cost can be reduced by providing the battery management system described above.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The accompanying drawings required for describing some embodiments will be introduced below, and the accompanying drawings described below are some accompanying drawings of the present application.

[0011] FIG. 1 is a schematic structural diagram of a battery pack according to some embodiments of the present application;

[0012] FIG. 2 is a sectional view of a battery pack according to some embodiments of the present application;

[0013] FIG. 3 is a schematic structural diagram of a battery management system according to some embodiments of the present application;

[0014] FIG. 4 is a sectional view of a battery management system according to some embodiments of the present application;

[0015] FIG. 5 is a schematic structural diagram of a fastening assembly according to some embodiments of the present application; and

[0016] FIG. 6 is a schematic structural diagram of an insulating member according to some embodiments of the present application, with a second fastener being omitted.

LIST OF REFERENCE NUMBERS

[0017] **100** battery management system [0018] **200** box body [0019] **300** battery module [0020] **201** first receiving cavity [0021] **202** second receiving cavity [0022] **10** first circuit board [0023] **11** pad [0024] **20** second circuit board [0025] **21** engaging hole [0026] **30** copper bar [0027] **40** fastening assembly [0028] **41** insulating member [0029] **411** plastic member [0030] **412** second fastener [0031] **413** receiving hole [0032] **414** snap [0033] **4141** engaging groove [0034] **415** limit groove [0035] **416** stiffener [0036] **42** first fastener

DETAILED DESCRIPTION

[0037] The present application will be described below in conjunction with the accompanying drawings and embodiments. The embodiments described herein are used to explain the present application. Some structures related to the present application are shown in the accompanying drawings for convenience of description.

[0038] In the description of the present application, unless otherwise expressly specified and limited, the term “connected to each other”, “connected”, or “fixed” is to be construed in a broad sense. For example, the term “connected” may refer to “fixedly connected” or “detachably connected” or “integrally connected”, may refer to “mechanically connected” or “electrically

connected”, or may refer to “connected directly”, “connected indirectly through an intermediary” or “internal connection between two elements, or interaction between two elements”. For those of ordinary skill in the art, meanings of the preceding terms in the present application may be understood based on situations.

[0039] In the present application, unless otherwise expressly specified and limited, a first feature being “on” or “under” a second feature may include the first feature and the second feature being in direct contact, or may include the first feature and the second feature not being in direct contact but being in contact with each other through an additional feature therebetween. Moreover, the first feature being “on”, “above” or “over” the second feature includes the first feature being directly on, above or over and obliquely on, above or over the second feature, or simply indicates that the first feature is at a higher level than the second feature. The first feature being “under”, “below” or “underneath” the second feature includes the first feature being directly under, below or underneath and obliquely under, below or underneath the second feature, or simply represents that the first feature is at a lower level than the second feature.

[0040] In the description of this embodiment, the orientational or positional relationships indicated by terms “above”, “below”, “left”, “right”, and the like are based on the orientational or positional relationships shown in the drawings, and are merely for ease of description and simplifying an operation, rather than indicating or implying that the referred device or element must have a specific orientation and be constructed and operated in a specific orientation. Moreover, the terms “first” and “second” are used only to distinguish between descriptions and have no special meaning.

[0041] This embodiment provides a battery management system **100** and a battery pack. As shown in FIGS. **1** and **2**, the battery pack includes a box body **200**, a battery module **300** and the battery management system **100** in any of the embodiments. The battery module **300** and the battery management system **100** are disposed within the box body **200**. In some embodiments, an interior space of the box body **200** is divided into a first receiving cavity **201** and a second receiving cavity **202**, the battery module **300** is disposed within the first receiving cavity **201**, the battery management system **100** is disposed within the second receiving cavity **202**, and the battery management system **100** is electrically connected to the battery module **300** to perform the intelligent management and maintenance on the battery module **300**.

[0042] In some embodiments, as shown in FIGS. **2** and **3**, the battery management system **100** includes a first circuit board **10**, a second circuit board **20**, a copper bar **30** and a fastening assembly **40**. The first circuit board **10** and the second circuit board **20** are arranged at intervals in a vertical direction and are fixed within the box body **200**. The copper bar **30** is located between the first circuit board **10** and the second circuit board **20**. The fastening assembly **40** includes an insulating member **41** and a first fastener **42**. The copper bar **30** is disposed to be attached to the first circuit board **10**. A first end of the insulating member **41** is connected to the second circuit board **20**, and a second end of the insulating member **41** supports the copper bar **30**. The first fastener **42** is fastenably connected to the insulating member **41** and abuts against a side of the first circuit board **10** away from the second circuit board **20**, to fix the copper bar **30** on the first circuit board **10**.

[0043] According to the battery management system **100** provided in this embodiment, the first circuit board **10** and the second circuit board **20** are arranged at intervals in a vertical direction and are fixed within the box body **200**, the copper bar **30** is supported by the insulating member **41** connected to the second circuit board **20**, and the fastener **42** is fastenably connected to the insulating member **41**, so that the copper bar **30** is attached and fixed on a side of the first circuit board **10** by using the pre-tightening force. The fastening assembly **40** of the above-described form is provided so that the copper bar **30** can be insulated from the box body **200** while the copper bar **30** is fastened, and the first fastener **42** is not in contact with the box body **200**; therefore, the copper bar **30** may not form an electrical connection with the box body **200** through the first

fastener **42**, thereby insulating and protecting the box body **200** without using the electrophoresis process, and reducing the manufacturing cost.

[0044] To fix the first circuit board **10** and the second circuit board **20** inside the box body **200**, in some embodiments, multiple connection columns are disposed at a cavity bottom of the second receiving cavity **202**, a threaded hole is disposed within each of the multiple connection columns, each of the first circuit board **10** and the second circuit board **20** is placed on a respective one among the multiple connection columns, and then a screw is disposed to penetrate through the first circuit board **10** or the second circuit board **20** and is in threaded connection with the threaded hole on each connection column of the multiple connection columns, so that each of the first circuit board **10** and the second circuit board **20** is fixed within the box body **200**.

[0045] In some embodiments, a first end of the copper bar **30** is attached and connected to the first circuit board **10** through the fastening assembly **40**, and a second end of the copper bar **30** is electrically connected to the battery module **300**. For an electrical connection manner between the copper bar **30** and the battery module **300**, reference may be made to the related art. As shown in FIGS. **2** and **3**, a side of the first circuit board **10** facing the copper bar **30** is provided with a pad **11**, and the copper bar **30** is attached to an exposed copper layer of the pad **11**, so that an electrical connection between the copper bar **30** and the first circuit board **10** is achieved. The copper bar **30** and the second circuit board **20** are disposed to be insulated from each other through the insulating member **41**. The first circuit board **10** may be electrically connected to the second circuit board **20** through a board end connector, a wire, a flexible printed circuit (FPC) and other structures. The number of copper bars **30** is set to two, two second ends of two copper bars **30** are electrically connected to the positive electrode and the negative electrode of the battery module **300**, respectively, and two first ends of the two copper bars **30** are fastenably connected to the first circuit board **10** through the fastening assembly **40** in the above-described embodiments. The first circuit board **10** is a power board and may pass a large current, and the second circuit board **20** may pass a low current. The current from the battery module **300** flows out through the copper bar **30**, and then is output to a powered device through the first circuit board **10**.

[0046] In an optional embodiment, after the first fastener **42** is disposed to penetrate through the first circuit board **10** and the copper bar **30** in sequence, the first fastener **42** is in threaded connection with the insulating member **41**. In some embodiments, as shown in FIGS. **4** and **5**, the insulating member **41** includes a plastic member **411** and a second fastener **412** embedded within the plastic member **411**. A first end of the plastic member **411** is fixed on the second circuit board **20**, and a second end of the plastic member **411** supports the copper bar **30**, so that an insulation connection between the copper bar **30** and the second circuit board **20** is achieved. A portion disposed to support the copper bar **30** of the plastic member **411** is a square structure, and another portion of the plastic member **411** is a cylindrical structure. A through hole for the first fastener **42** to penetrate through is disposed on the first circuit board **10** and the copper bar **30**, and the first fastener **42** is disposed to penetrate through the first circuit board **10** and the copper bar **30** in sequence and then is in threaded connection with the second fastener **412** to satisfy the pre-tightening function. The first fastener **42** is screwed so that a head of the first fastener **42** abuts against the side of the first circuit board **10** away from the second circuit board **20**. A side of the copper bar **30** is tightly attached to a side of the first circuit board **10** facing the second circuit board **20**, and another side of the copper bar **30** is supported by an end of the plastic member **411**, whereby the fastening effect of the copper bar **30** is achieved. In this embodiment, the first fastener **42** is a screw, the second fastener **412** is a nut, a threaded rod portion of the screw is disposed to penetrate through the first circuit board **10** and the copper bar **30** in sequence and then is in threaded connection with the nut, and a head of the screw abuts against the side of the first circuit board **10** away from the second circuit board **20**.

[0047] In another optional embodiment, a hole site may be reserved on the plastic member **411**, and then a threaded hole may be formed by tapping in the hole site. The first fastener **42** may be a

screw. A threaded rod portion of the screw is disposed to penetrate through the first circuit board **10** and the copper bar **30** in sequence and is in threaded connection with the threaded hole on the plastic member **411**, and a head portion of the screw abuts against on the side of the first circuit board **10** away from the second circuit board **20**, so that the fastening effect of the copper bar **30** can also be achieved. In this implementation, the second fastener **412** does not need to be embedded within the plastic member **411**.

[0048] In another optional embodiment, the first fastener **42** may also be selected as a self-tapping screw. After the self-tapping screw is disposed to penetrate through the first circuit board **10** and the copper bar **30** in sequence, the self-tapping screw is tightened directly by using a tool, so that threads can be automatically formed in the reserved hole of the plastic member **411** during rotation, whereby the fastening effect of the copper bar **30** can also be achieved without the aid of the nut or the pre-tapping. Moreover, in this implementation, the second fastener **412** does not need to be embedded within the plastic member **411**.

[0049] In another optional embodiment, the insulating member **41** is disposed to penetrate through the copper bar **30** and the first circuit board **10** in sequence, and then is in threaded connection with the first fastener **42**. In some embodiments, the insulating member **41** includes a plastic member **411** and a second fastener **412** embedded within the plastic member **411**, and the second fastener **412** is disposed to penetrate through the copper bar **30** and the first circuit board **10** in sequence and then is in threaded connection with the first fastener **42** to satisfy the pre-tightening function. In some embodiments, the first fastener **42** may be the nut, the second fastener **412** may be the bolt. The bolt is embedded within the plastic member **411**. During mounting, a stud portion of the bolt is disposed to penetrate through the copper bar **30** and the first circuit board **10** in sequence, then the nut is screwed onto the bolt, and the nut abuts against the side of the first circuit board **10** away from the second circuit board **20**, so that the fastening effect of the copper bar **30** can also be achieved.

[0050] As shown in FIGS. **4** and **6**, in this embodiment, a receiving hole **413** is disposed within the plastic member **411**, and the second fastener **412** is embedded within the receiving hole **413**, and the receiving hole **413** is a blind hole. The first fastener **42** in this embodiment is the screw and the second fastener **412** in this embodiment is the nut; therefore metal debris is generated within the nut when the screw is screwed, and moreover, the receiving hole **413** is disposed to be the blind hole, so that the metal debris can be directly dropped into the blind hole. Compared with a setting that the receiving hole **413** is the through hole, a phenomenon that the metal debris is attached to the second circuit board **20** to cause the short circuit can be avoided.

[0051] As shown in FIGS. **3** and **4**, an engaging hole **21** is disposed on the second circuit board **20**, a snap **414** is disposed on a circumferential side of the plastic member **411**, and the snap **414** is engaged within the engaging hole **21**. The snap **414** and the plastic member **411** are of an integrally formed structure, and the snap **414** may be assembled with the engaging hole **21** through the elastic deformation of the snap **414** itself and is finally engaged within the engaging hole **21**, so that the plastic member **411** is firmly connected to the second circuit board **20**, and thus the assembling and disassembling are relatively convenient.

[0052] In some embodiments, as shown in FIGS. **4** and **5**, a side of the snap **414** facing the plastic member **411** is bent to form an engaging groove **4141**. After the snap **414** passes through the engaging hole **21**, a side edge of the second circuit board **20** is engaged within the engaging groove **4141**, so that a stable connection between the plastic member **411** and the second circuit board **20** is achieved, and the disassembly and assembly is convenient.

[0053] In some embodiments, the mounting process of the battery management system **100** is as follows. The snap **414** on the circumferential side of the plastic member **411** is firstly engaged into the engaging hole **21** of the second circuit board **20**, so that the plastic member **411** and the second circuit board **20** are assembled into one piece, where the plastic member **411** is the plastic member **411** into which the second fastener **412** has been embedded. After the assembly, the plastic member

411 and the second circuit board **20** are integrated into the box, the second circuit board **20** is locked within the box body **200** through screws, then the copper bar **30** is placed on the plastic member **411**, and then the first circuit board **10** is placed on the copper bar **30**, so that the first circuit board **10** tightly presses the copper bar **30**. In this case, the through hole on the first circuit board **10** is aligned with the through hole on the copper bar **30**, the first fastener **42** is disposed to penetrate through the through hole of the first circuit board **10** and the through hole of the copper bar **30** in sequence and is then in threaded connection with the second fastener **412** embedded within the plastic member **411** to lock the first fastener **42**, whereby the copper bar **30** is attached and fixed on the first circuit board **10**, and finally, the first circuit board **10** is locked within the box body **200** by screws. After the battery management system **100** is assembled, the whole box body **200** is turned over, so that an opening of the first receiving cavity **201** faces upwards, and then the battery module **300** is mounted.

[0054] As shown in FIG. 5, the snap **414** is widened in an X-axis direction. The snap **414** is widened, so that the structural strength of the snap **414** can be enhanced, and the snap **414** is prevented from being deformed or damaged when the first fastener **42** is screwed, thereby preventing the plastic member **411** from following the rotation.

[0055] Referring to FIG. 4 to FIG. 6, an end of the plastic member **411** facing the copper bar **30** is provided with a limit groove **415**, and the copper bar **30** is accommodated within the limit groove **415**. The limit groove **415** is configured to limit the copper bar **30**, and after the copper bar **30** is placed within the limit groove **415**, the copper bar **30** can be prevented from shaking and shifting in a Y direction.

[0056] As shown in FIGS. 5 and 6, a stiffener **416** is disposed on a circumferential side wall of the plastic member **411**, the stiffener **416** is connected between a top side of the snap **414** and a bottom side of the square structure of the plastic member **411**, and the plastic member **411**, the stiffener **416** and the snap **414** are all of an integrally formed structure. The stiffener **416** is provided so that the overall structural strength of the plastic member **411** can be improved to prevent the snap **414** from being deformed or damaged when the first fastener **42** is screwed.

Claims

1. A battery management system, disposed within a box body, wherein the battery management system comprises: a first circuit board and a second circuit board which are arranged at intervals and fixed within the box body; a copper bar located between the first circuit board and the second circuit board; and a fastening assembly, comprising an insulating member and a first fastener, wherein the copper bar is disposed to be attached to the first circuit board, a first end of the insulating member is connected to the second circuit board, a second end of the insulating member supports the copper bar, and the first fastener is fastenably connected to the insulating member and abuts against a side of the first circuit board away from the second circuit board, to fix the copper bar on the first circuit board.
2. The battery management system of claim 1, wherein the first fastener is disposed to penetrate through the first circuit board and the copper bar in sequence and is then in threaded connection with the insulating member; or the insulating member is disposed to penetrate through the copper bar and the first circuit board in sequence and is then in threaded connection with the first fastener.
3. The battery management system of claim 1, wherein the insulating member comprises a plastic member and a second fastener embedded within the plastic member, a first end of the plastic member is fixed on the second circuit board, a second end of the plastic member supports the copper bar, and the first fastener is connected to the second fastener.
4. The battery management system of claim 3, wherein the first fastener is a screw, the second fastener is a nut, and the screw is disposed to penetrate through the first circuit board and the copper bar in sequence and is in threaded connection with the nut.

5. The battery management system of claim 3, wherein a receiving hole is disposed within the plastic member, the second fastener is embedded within the receiving hole, and the receiving hole is a blind hole.
 6. The battery management system of claim 1, wherein an engaging hole is disposed on the second circuit board, a circumferential side of the insulating member is provided with a snap, and the snap is engaged into the engaging hole.
 7. The battery management system of claim 6, wherein an engaging groove is formed on the snap, and after the snap penetrates through the engaging hole, a side edge of the second circuit board is engaged within the engaging groove.
 8. The battery management system of claim 1, wherein an end of the insulating member facing the copper bar is provided with a limit groove, and the copper bar is accommodated within the limit groove.
 9. The battery management system of claim 1, wherein a stiffener is disposed on a circumferential side wall of the insulating member.
 10. The battery management system of claim 2, wherein an engaging hole is disposed on the second circuit board, a circumferential side of the insulating member is provided with a snap, and the snap is engaged into the engaging hole.
 11. The battery management system of claim 3, wherein an engaging hole is disposed on the second circuit board, a circumferential side of the insulating member is provided with a snap, and the snap is engaged into the engaging hole.
 12. The battery management system of claim 4, wherein an engaging hole is disposed on the second circuit board, a circumferential side of the insulating member is provided with a snap, and the snap is engaged into the engaging hole.
 13. The battery management system of claim 5, wherein an engaging hole is disposed on the second circuit board, a circumferential side of the insulating member is provided with a snap, and the snap is engaged into the engaging hole.
 14. The battery management system of claim 2, wherein an end of the insulating member facing the copper bar is provided with a limit groove, and the copper bar is accommodated within the limit groove.
 15. The battery management system of claim 3, wherein an end of the insulating member facing the copper bar is provided with a limit groove, and the copper bar is accommodated within the limit groove.
 16. The battery management system of claim 4, wherein an end of the insulating member facing the copper bar is provided with a limit groove, and the copper bar is accommodated within the limit groove.
 17. The battery management system of claim 5, wherein an end of the insulating member facing the copper bar is provided with a limit groove, and the copper bar is accommodated within the limit groove.
 18. The battery management system of claim 2, wherein a stiffener is disposed on a circumferential side wall of the insulating member.
 19. The battery management system of claim 3, wherein a stiffener is disposed on a circumferential side wall of the insulating member.
 20. A battery pack, comprising a box body, a battery module and the battery management system of claim 1, wherein the battery module and the battery management system are disposed within the box body, and the battery module is electrically connected to the battery management system.
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